

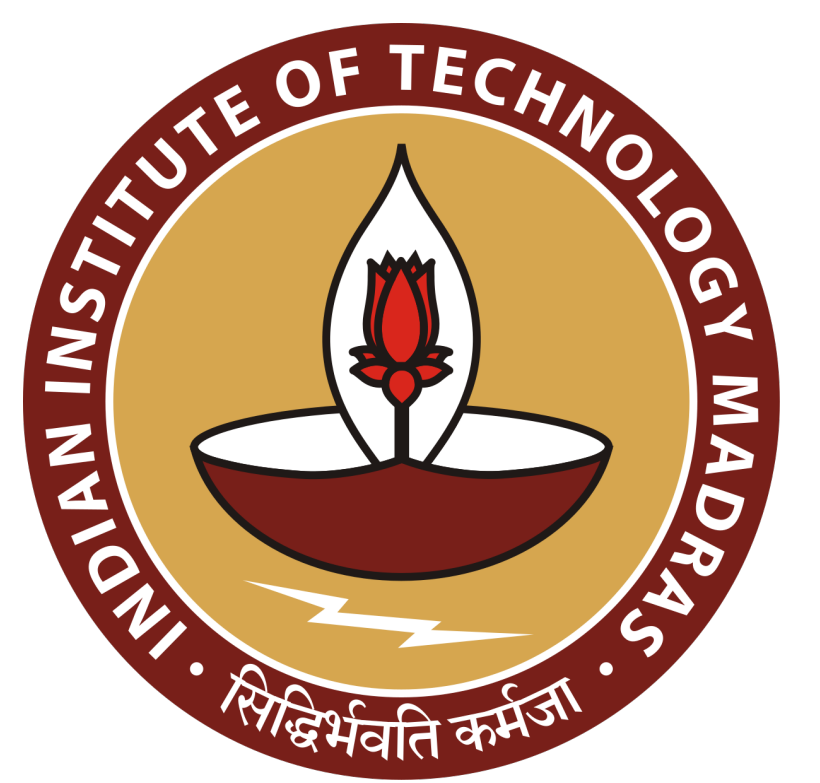
ON EIGENVALUES OF LOGARITHMIC POTENTIAL OPERATOR

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Logarithmic potential operator ↓

1. Definition

For a bounded open set $\Omega \subset \mathbb{R}^2$, the logarithmic potential operator $\mathcal{L}$ on $L^2(\Omega)$:

$$\mathcal{L}u(x) = \int_{\Omega} \log \frac{1}{|x-y|} u(y) dy$$

- $\frac{1}{2\pi} \log \frac{1}{|x|} \rightarrow$ the fundamental solution of Laplacian in $\mathbb{R}^2$

Poisson's equation $\rightarrow -\Delta w = u$

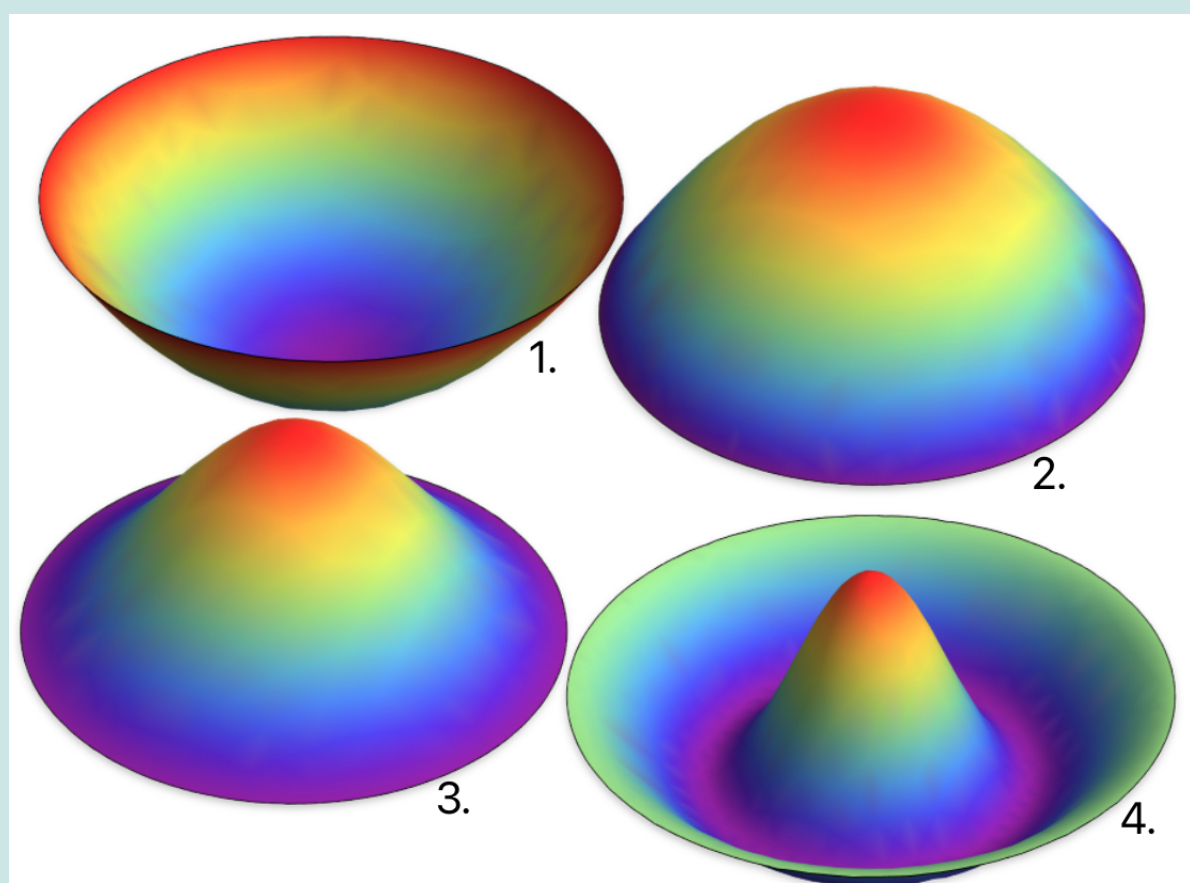
$$w(x) = \frac{1}{2\pi} \int_{\Omega} \log \frac{1}{|x-y|} u(y) dy = \frac{1}{2\pi} \mathcal{L}u(x)$$

$$\Rightarrow -\Delta \mathcal{L}u = 2\pi u$$

- $\mathcal{L}$ is not a positive operator in general

2. Eigenfunctions

- An eigenfunction corresponding to a negative eigenvalue has a constant sign
- If $T_{diam}(\bar{\Omega}) > 1$, there is no positive eigenfunction corresponding to the largest eigenvalue



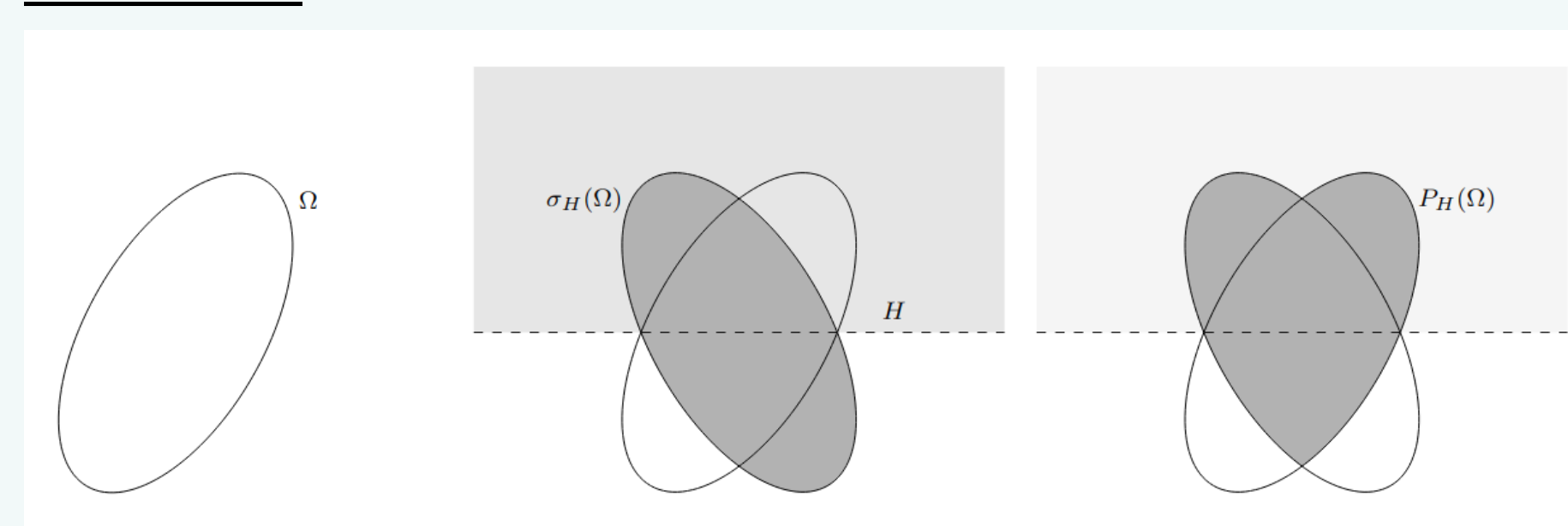
Eigenfunctions on disks: 1. Positive for the unique negative eigenvalue on B_e ; 2. Positive for the largest eigenvalue on $B_{1/e}$; 3 & 4. Sign-changing radial eigenfunctions on B_e .

3. Polarization

For $\Omega \subset \mathbb{R}^N$, the polarization w.r.t a polarizer H :

$$P_H(\Omega) = [(\Omega \cup \sigma_H(\Omega)) \cap H] \cup [\Omega \cap \sigma_H(\Omega)]$$

Example:



4. Reverse Faber-Krahn inequalities

- $\tau^+ \rightarrow$ the largest eigenvalue of $\mathcal{L}$

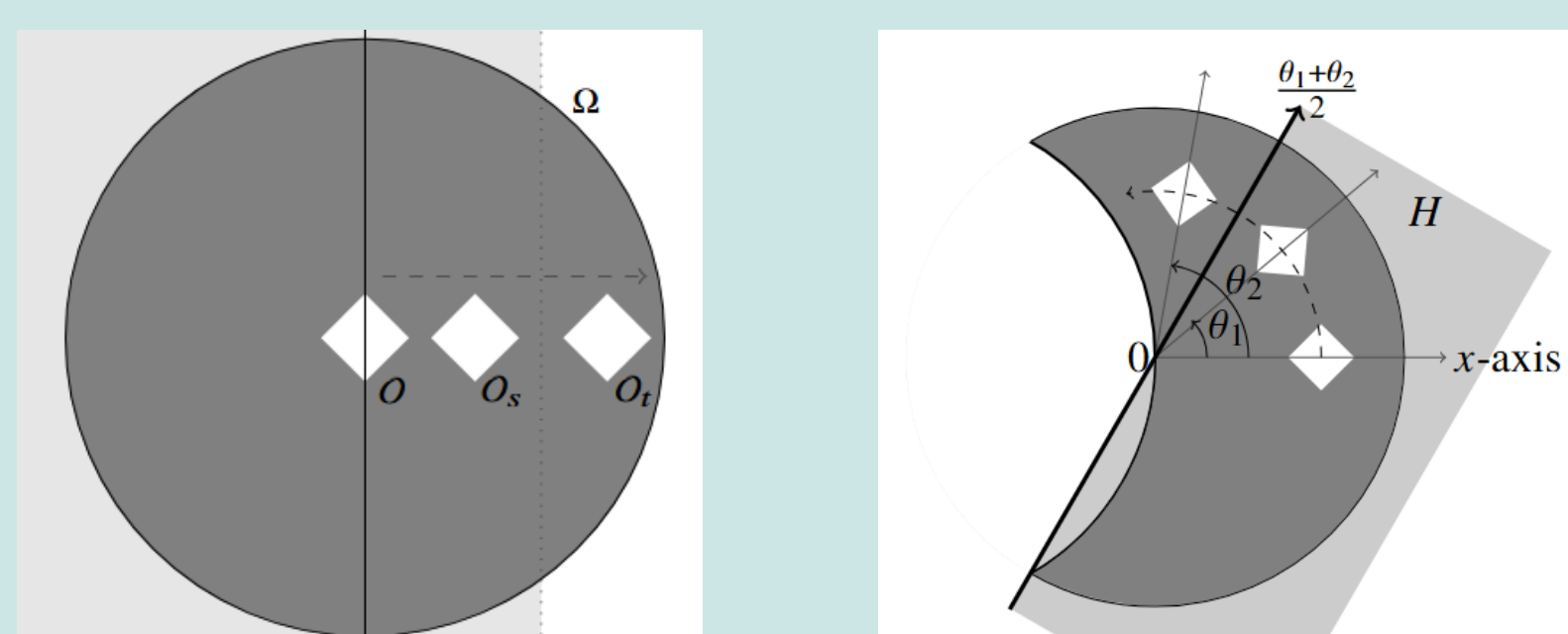
Let $\Omega \subset \mathbb{R}^2$ be such that an eigenfunction corresponding to $\tau^+(\Omega)$ is positive on Ω . Then,

$$\tau^+(\Omega) \leq \tau^+(P_H(\Omega)),$$

and

$$\tau^+(\Omega) \leq \tau^+(\Omega^*).$$

Domain perturbations



$$\tau^+(\Omega \setminus O_s) < \tau^+(\Omega \setminus O_l)$$

$$\tau^+(\Omega \setminus O_{\theta_1}) < \tau^+(\Omega \setminus O_{\theta_2})$$

Weighted logarithmic operator ↓

5. Definition

Ω bounded open subset of $\mathbb{R}^N$

$w, g \in C^+(\bar{\Omega})$, positive continuous functions

The weighted logarithmic potential operator $\mathcal{L}_w$ on $L^2(\Omega)$:

$$\mathcal{L}_w u(x) = \int_{\Omega} \log \left(\frac{w(x)w(y)}{|x-y|} \right) u(y) dy$$

The weighted eigenvalue problem:

$$\mathcal{L}_w u = \tau g u$$

6. Weighted transfinite diameter

Let E be a compact set in $\mathbb{R}^N$ and $W \in C^+(E)$. For each $n \geq 2$, the n^{th} -weighted diameter of E :

$$\rho_n(E, W) = \max_{(x_1, x_2, \dots, x_n) \in E^n} \left[\prod_{1 \leq i < j \leq n} |x_i - x_j| W(x_i) W(x_j) \right]^{\frac{2}{n(n-1)}}$$

The weighted transfinite diameter of E :

$$T_{diam}(E, W) = \lim_{n \rightarrow \infty} \rho_n(E, W)$$

7. Eigenvalues

There exists a sequence of nonzero eigenvalues with corresponding eigenfunctions in $C(\bar{\Omega})$.

Existence of a negative eigenvalue:

$$\begin{aligned} \log w \text{ is superharmonic} &\Rightarrow \mathcal{L}_w \text{ admits a negative eigenvalue} \\ T_{diam}(\bar{\Omega}, \frac{1}{w}) &\text{ is bigger than 1} \end{aligned}$$

Can zero be an eigenvalue?

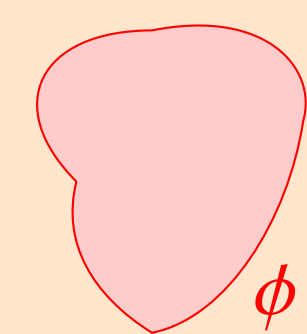
If $\Delta(\log w) = C \neq \frac{2\pi}{|\Omega|}$, then 0 is not an eigenvalue.

$$\begin{aligned} \text{On a disc } B_R, \text{ consider } w_1(x) &= e^{\frac{|x|^2}{2R^2} + \frac{1}{2} \log R - \frac{3}{8}} \\ \text{For } w &= c w_1, c \neq 1, \text{ No!} \\ \text{For } w &= w_1, \text{ Yes!} \end{aligned}$$

8. Geometry of eigenfunctions

Assume $\log w$ is harmonic and $g = 1$

$$\begin{aligned} \tau > 0 &\Rightarrow \text{for } \phi \rightarrow \begin{cases} \text{no positive local minimum} \\ \text{no negative local maximum} \end{cases} \\ \tau < 0 &\Rightarrow \text{for } \phi \rightarrow \begin{cases} \text{no negative local minimum} \\ \text{no positive local maximum} \end{cases} \end{aligned}$$



$$\phi \geq 0 \text{ on } \partial\Omega$$

$$\Rightarrow \phi \geq 0 \text{ on } \Omega$$

(Maximum principle)

9. Reverse Faber-Krahn inequality

$$\tau_{w,g}^+(\Omega) = \max_{\substack{u \in L^2(\Omega) \\ u \neq 0}} \frac{\langle \mathcal{L}_w u, u \rangle}{\int_{\Omega} g u^2}$$

$w(x) = w(\bar{x})$ and $g(x) = g(\bar{x})$

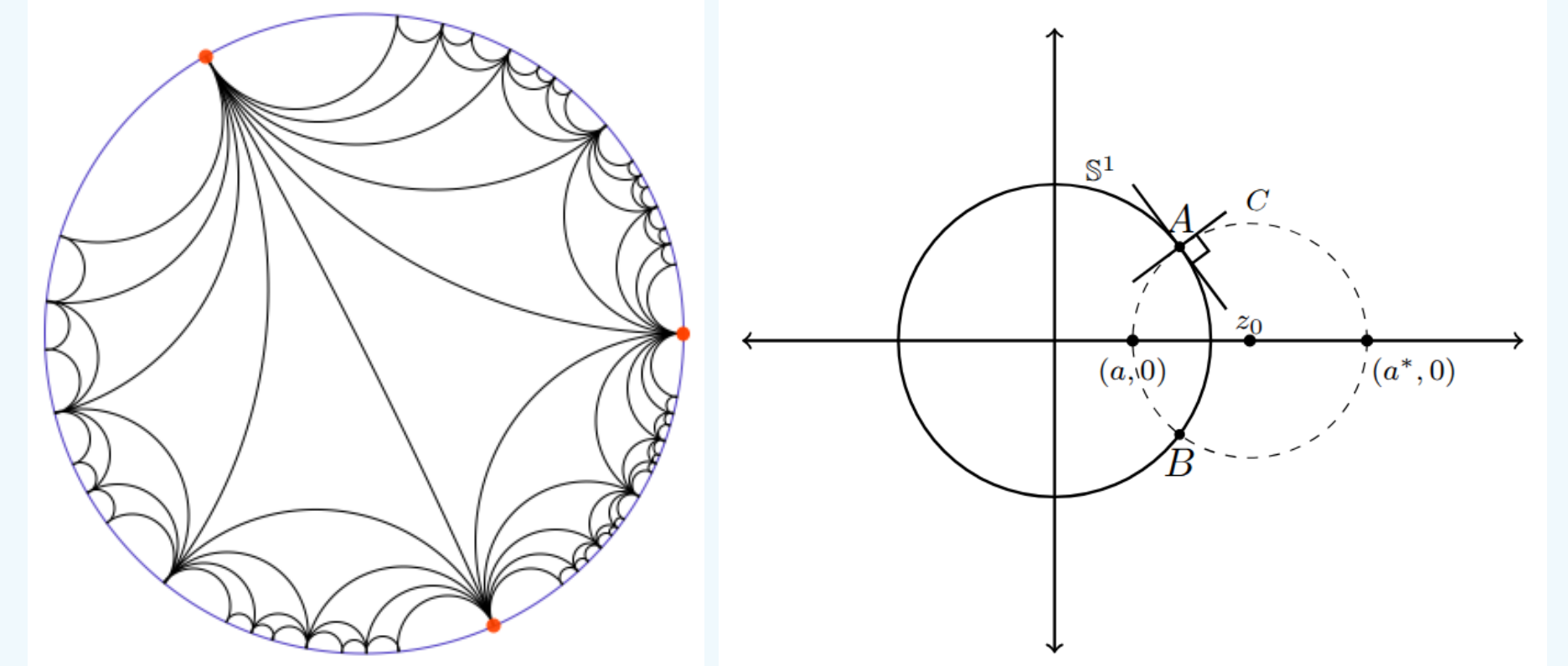
an eigenfunction corresponding to $\tau_{w,g}^+(\Omega)$ is positive

$$\tau_{w,g}^+(\Omega) \leq \tau_{w,g}^+(P_H(\Omega))$$

Hyperbolic logarithmic operator ↓

10. Definition

Poincaré disk model:



$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$$

The hyperbolic metric:

$$d_h(z, w) = 2 \tanh^{-1} \left| \frac{z-w}{1-\bar{z}w} \right|$$

The pseudo-hyperbolic metric:

$$[z, w] = \left| \frac{z-w}{1-\bar{z}w} \right|$$

Definition of the operator:

Let $\Omega \subset \mathbb{D}$ be an open set, bounded with respect to the hyperbolic metric d_h . The hyperbolic logarithmic operator $\mathcal{L}_h$ on $L^2(\Omega)$:

$$\mathcal{L}_h(u)(z) = \int_{\Omega} \log \frac{1}{[z, w]} u(w) d(w),$$

where

$$d(w) = \frac{1}{\pi} \frac{dA(w)}{(1-|w|^2)^2}.$$

$\Gamma_h(x) = \frac{1}{2} \log \frac{1}{|x|} \rightarrow$ the fundamental solution of Δ_h

Invariant convolution:

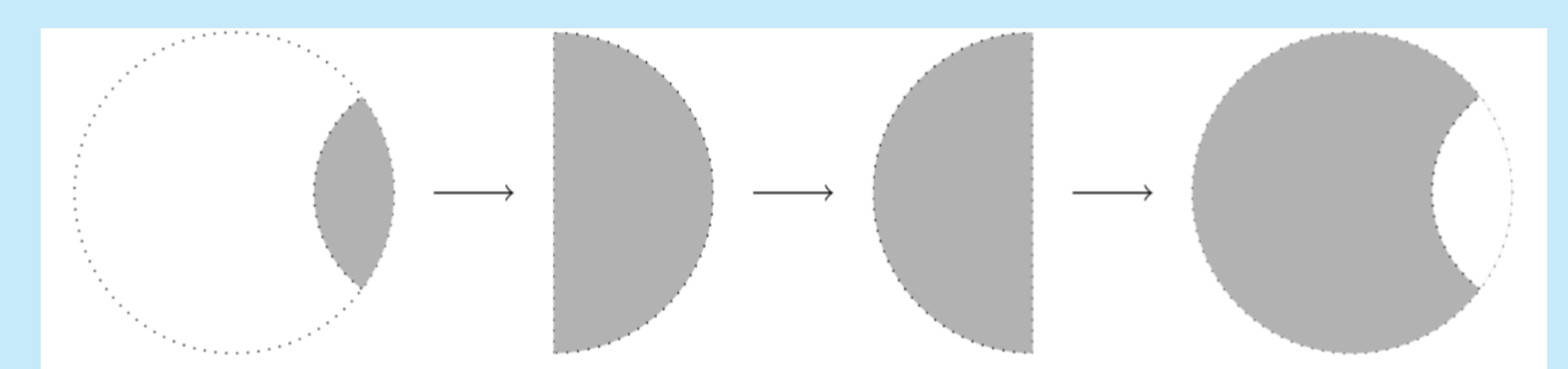
$$(f \star g)(z) = \int_{\mathbb{D}} f(w) g\left(\frac{z-w}{1-\bar{z}w}\right) d(w)$$

Under suitable assumptions on u , $\mathcal{L}_h(u) = u \star \Gamma_h$ solves the equation $-\Delta_h w = 2u$

11. All eigenvalues of $\mathcal{L}_h$ are positive

- $\mathcal{L}_h$ is a positive operator on $L^2(\Omega)$
- $\mathcal{L}_h$ does not admit negative eigenvalues
- Zero is not an eigenvalue of $\mathcal{L}_h$

12. Reflection and polarization



The reflection with respect to a geodesic $\mathcal{G}$:

$$z \rightarrow T(z) \rightarrow \overline{-T(z)} \rightarrow T^{-1}(\overline{-T(z)})$$

$$\sigma_{\mathcal{G}}(z) = T^{-1}(\overline{-T(z)})$$

For a set $\Omega \subset \mathbb{D}$, the polarization of Ω with respect to a polarizer $\mathcal{H}$:

$$P_{\mathcal{H}}(\Omega) = [(\Omega \cup \sigma_{\mathcal{G}}(\Omega)) \cap \mathcal{H}] \cup [\Omega \cap \sigma_{\mathcal{G}}(\Omega)]$$

13. Reverse Faber-Krahn inequality

- $\tau_h \rightarrow$ the largest eigenvalue of $\mathcal{L}_h$
- $\mathcal{G}$ is a geodesic and $\mathcal{H}$ is an associated polarizer

$$\tau_h(\Omega) \leq \tau_h(P_{\mathcal{H}}(\Omega))$$

References

- [1] T. V. Anoop and Jiya Rose Johnson. "Reverse Faber-Krahn inequalities for the logarithmic potential operator". In: *Discrete and Continuous Dynamical Systems* (2026).
- [2] T. V. Anoop and Jiya Rose Johnson. *On the weighted logarithmic potential operator*. 2026. arXiv: 2602.18138.
- [3] Jiya Rose Johnson and Sheela Verma. *On eigenvalues of logarithmic potential operator in the hyperbolic space*. 2026. arXiv: 2601.20431.



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